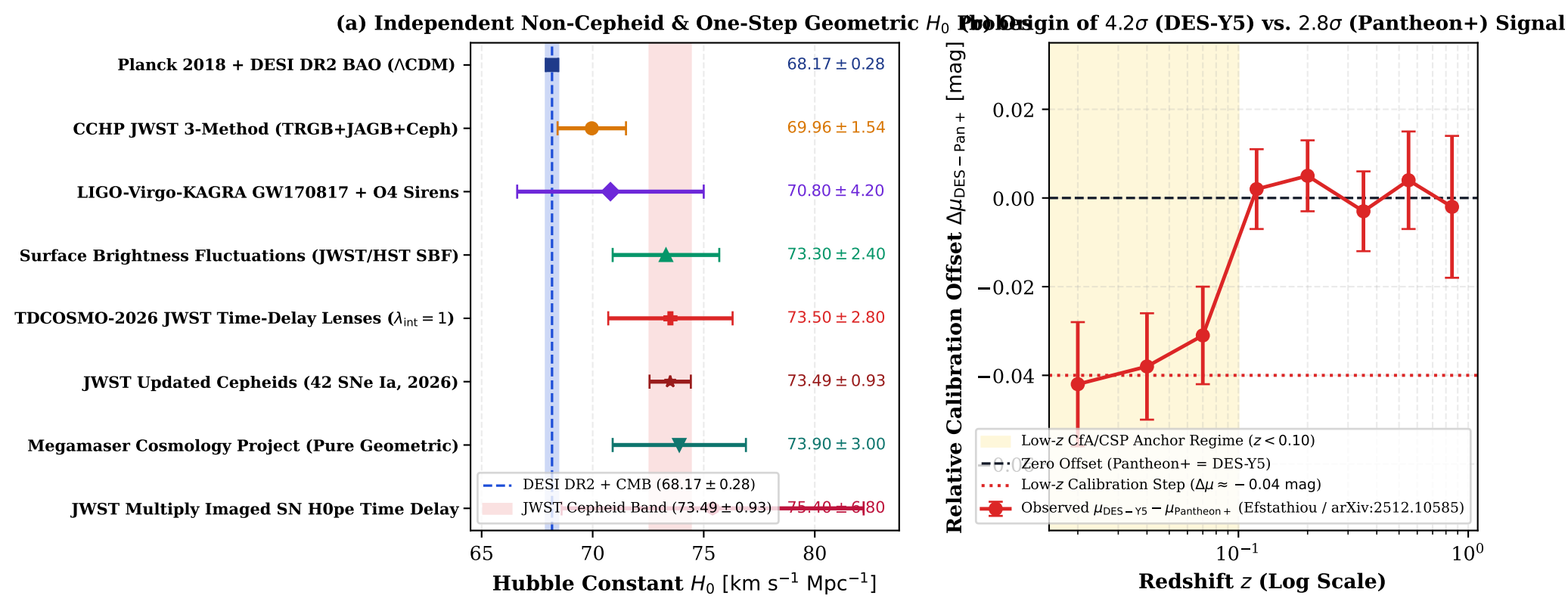

(a) Independent Non-Cepheid & One-Step Geometric H_0 Probes
Planck 2018 + DESI DR2 BAO (ΛCDM)
68.17 ± 0.28
CCHP JWST 3-Method (TRGB+JAGB+Ceph)
69.96 ± 1.54
LIGO-Virgo-KAGRA GW170817 + O4 Sirens
70.80 ± 4.20
Surface Brightness Fluctuations (JWST/HST SBF)
73.30 ± 2.40
TDCOSMO-2026 JWST Time-Delay Lenses ($\lambda_{int} = 1$)
73.50 ± 2.80
JWST Updated Cepheids (42 SNe Ia, 2026)
73.49 ± 0.93
Megamaser Cosmology Project (Pure Geometric)
73.90 ± 3.00
JWST Multiply Imaged SN H0pe Time Delay
75.40 ± 6.80
DESI DR2 + CMB (68.17 ± 0.28)
JWST Cepheid Band (73.49 ± 0.93)
65
70
75
80
Hubble Constant H_0 [km s^{-1} Mpc^{-1}]
(b) Origin of 4.2σ (DES-Y5) vs. 2.8σ (Pantheon+) Signal
Relative Calibration Offset $\Delta\mu_{DES-Pan+}$ [mag]
0.02
0.00
−0.02
−0.04
Low-z CfA/CSP Anchor Regime (z < 0.10)
Zero Offset (Pantheon+ = DES-Y5)
Low-z Calibration Step (Δμ ≈ −0.04 mag)
Observed $\mu_{DES-Y5} - \mu_{Pantheon+}$ (Efstathiou / arXiv:2512.10585)
10^{-1}
10^{0}
Redshift z (Log Scale)